Finlay Lemon

Mechatronics and Robotics Engineering student, Queen's University | Available from Jan. 2027 | Willing to relocate

Education

Queen's University Sep. 2024 – Apr. 2028
Mechatronics and Robotics Engineering, BASc *Kingston, Ontario*

- Relevant coursework: Thermodynamics, Fluid Mechanics, Solid Mechanics

Projects

Lunar Logistics Rover | *SolidWorks, SolidWorks Simulation* Aug. 2026 – Present

- Developing an independent concept design aimed at the Canadian Space Agency's Lunar Utility Vehicle (LUV) program, maintaining interface control across electrical, mechanical, and thermal subsystems.
- Defined the thermal control architecture: a dual-mode, mechanically pumped fluid loop that uses RTG waste heat to keep cold components warm and carries heat away from the six drive units and all electronics.
- Applied Bekker terramechanics to develop a novel wheel concept and validated it with FEA, raising flexure FoS from below 1.0 to 1.68 without significant mass increase.
- Documented the design in a consolidated baseline, trade studies, and a multiple-requirement specification register, updated as the design changes.

Room-mapping Rover | *ROS 2, RPLidar, Python, RViz, Gazebo* Jan. 2026 – Apr. 2026

- Built an autonomous rover on the Lynxmotion 4WD platform in a team of 3, using an RPLidar for LiDAR-based SLAM mapping in ROS 2, with RViz for visualization.
- Integrated the SLAM mapping and developed the point-to-point navigation; used Gazebo Sim as a proof-of-concept before testing on the physical rover.

Desktop Wind Tunnel | *SolidWorks, FDM 3D Printing* Oct. 2025 – Nov. 2025

- Designed and built a working, mostly 3D-printed benchtop wind tunnel with streamline visualization, as a low-cost alternative to the large commercial units used at Queen's.
- Used a modular design so it can be made in about 20 hours with household filament; tested the airflow and streamline visualization to evaluate the completed design.

Experience

Queen's Rocket Engineering Team (QRET) 2024 – Present
Research Lead (Sep. 2025 – Apr. 2026); Mechanical Team Member *Kingston, Ontario*

- Designing the mechanical interface between the avionics circuits and the airframe of a hybrid-propulsion supersonic sounding rocket, including an access door through the airframe.
- Co-led a 15-person team to deliver a complete CAN-RGX microgravity research proposal on schedule within a 2-month window, coordinating sourcing and internal review cycles.
- Built the proposal's science case around triply periodic minimal surface (TPMS) high-surface-area heat exchanger structures that transfer heat without relying on convection in microgravity.

Technical Skills

CAD & Design: SolidWorks (4 years), Siemens NX (6 months), FDM 3D printing
Analysis: SolidWorks Simulation (FEA), MATLAB / Simulink, Python, Excel
Robotics & Software: ROS 2, RViz, Gazebo
Electronics: KiCad, LTspice, soldering, multimeter/oscilloscope
Hardware: LiDAR (RPLidar), mechanical assembly, data collection
Certifications: Amateur Radio — ISED Canada 'Advanced' Qualification